

Scaling Book — Section 2 Exercises

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Q1

Sample from a 200B-parameter model in bf16. Split weights across 32 TPU v4p.

Per TPU:

$$\frac{2 \cdot 200 \times 10^9 \text{ bytes}}{32} = 1.25 \times 10^{10} \text{ bytes/TPU.}$$

TPU v4p HBM $\rightarrow$ VMEM bandwidth = 1.2e12 bytes/s.

Total time to load to systolic array:

$$\frac{1.25 \times 10^{10}}{1.2 \times 10^{12}} \text{ s} \approx 0.01 \text{ s} = 10 \text{ ms.}$$

If sampling from the model is memory bound, then this becomes a lower bound for the latency of sampling.

Q2

(a) TPU v5e pod (16×16 2D torus)

- (i) Each host is connected to 2 four-chip trays:

$$\text{total hosts} = \frac{16 \cdot 16}{8} = 32 \text{ cpu hosts.}$$

- (ii) Each v5e chip has 1 TPU core $\Rightarrow$ 256 total TPU cores in the v5e pod.

- (iii) v5e FLOPs/s/chip (bf16) = 1.97×10^{14} :

$$\text{total pod FLOPs/s} = 256 \cdot 1.97 \times 10^{14} \approx 5.04 \times 10^{16} \text{ FLOPs/s.}$$

- (iv) Total HBM = $256 \times 16 \text{ GB} = 4096 \text{ GB}$.

(b) TPU v5p pod ($16 \times 20 \times 28$ 3D torus)

- (i) One cpu host is connected to one 4-chip tray:

$$\text{total hosts} = \frac{16 \times 20 \times 28}{4} = 2240 \text{ cpu hosts.}$$

(ii) Each v5p chip has 2 TPU cores:

$$\text{total cores} = 2 \cdot (16 \times 20 \times 28) = 17,920 \text{ TPU cores.}$$

(iii) Total FLOPs/s (bf16):

$$(16 \times 20 \times 28) \cdot 4.59 \times 10^{14} \approx 4.1 \times 10^{18} \text{ FLOPs/s.}$$

(iv) Total HBM = $(16 \times 20 \times 28) \cdot 96 \text{ GB} = 860,160 \text{ GB} \approx 860 \text{ TB}$.

Q3

$$A : \text{bf16}[D, F], \quad X : \text{bf16}[B, D], \quad X \cdot A : \text{bf16}[B, F].$$

Assume $B \ll D$, $F = 4D$, on TPU v6e.

$$\text{total FLOPs} = BF(D + D - 1) = 2BFD$$

$$\text{bytes moved} = 2DF + 2BD + 2BF$$

Intensity (accel) — over PCIe:

$$\text{Intensity(accel)} = \frac{9.20 \times 10^{14}}{1.6 \times 10^{10}} = 57,500.$$

Intensity (comp):

$$\begin{aligned} \frac{2BFD}{2DF + 2BD + 2BF} &\xrightarrow{F=4D} \frac{2B \cdot 4D \cdot D}{2 \cdot 4D \cdot D + 2BD + 2B \cdot 4D} \\ &= \frac{8BD^2}{8D^2 + 2BD + 8BD} \\ &= \frac{8BD^2}{8D^2 + 10BD} \xrightarrow{B \ll D} B. \end{aligned}$$

$\therefore$ need $B \geq 57,500$ for the computation to remain FLOPs-bound over PCIe.

Q4

$$W : \text{int8}[16384, 4096], \quad x : \text{int8}[B, 4096], \quad x \cdot W \rightarrow \text{int8}[B, 16384] \quad \text{on 1 TPU v5e.}$$

Common quantities:

$$\text{FLOPs} = 16384 \cdot B \cdot (2 \cdot 4096) = 134,217,728 B \approx 1.34 \times 10^8 B$$

$$\text{bytes} = (16384 \cdot 4096) + (B \cdot 4096) + (B \cdot 16384)$$

$$= 20,480 B + 6.7 \times 10^7$$

(i) HBM

$$T_{\text{math}} = \frac{\text{FLOPs}}{\text{v5e FLOPs/s}} = \frac{1.34 \times 10^8 B}{3.94 \times 10^{14}} \approx 3.4 \times 10^{-7} B$$

$$T_{\text{mem}} = \frac{\text{bytes}}{\text{v5e bw}} = \frac{20,480 B + 6.7 \times 10^7}{8.2 \times 10^{11}}$$

$$\text{total time} \geq \max\{T_{\text{math}}, T_{\text{mem}}\}.$$

Compute-bound threshold^{††} *Continuation from page 4 of the notes.*

$$\text{Intensity}(\text{comp}) = \frac{1.34 \times 10^8 B}{2 \times 10^4 B + 6.7 \times 10^7}$$

$$\text{Intensity}(\text{accel}) = \frac{3.94 \times 10^{14}}{8.2 \times 10^{11}} \approx 480$$

To be compute-bound, $\text{Intensity}(\text{comp}) \geq \text{Intensity}(\text{accel})$:

$$(1.34 \times 10^8) B \geq 480 \cdot (2 \times 10^4 B + 6.7 \times 10^7)$$

$$(1.34 \times 10^8) B \geq (9.6 \times 10^6) B + 480 \cdot 6.7 \times 10^7$$

$$(1.34 \times 10^8 - 9.6 \times 10^6) B \geq 480 \cdot 6.7 \times 10^7$$

$$B \geq \frac{480 \cdot 6.7 \times 10^7}{1.34 \times 10^8 - 9.6 \times 10^6} \approx 259.$$

(ii) VMEM only

VMEM bandwidth $\approx 22 \times$ HBM bandwidth, so

$$\text{total time} \geq \max \left\{ \underbrace{3.4 \times 10^{-7} B}_{T_{\text{math}} \text{ (stays same)}}, \underbrace{\frac{2 \times 10^4 B + 6.7 \times 10^7}{22 \cdot 8.2 \times 10^{11}}}_{T_{\text{mem}} \text{ (22} \times \text{ lower)}} \right\}.$$

Compute-bound threshold^{††} *Continuation from page 4 of the notes.*

Using VMEM only, $\text{Intensity}(\text{accel})$ drops by $22 \times$, so

$$B \geq \frac{259}{22} \approx 11.$$

Q5

TPU v5e 4×4 slice, per-hop latency is $1 \mu\text{s}$. Want to send bf16[8, 128, 8192] from TPU{0, 0} to TPU{3, 3}.

Full v5e pod is 16×16 . No wrap around connection in slice!

Note: we have 2 output connections from TPU{0,0} and 2 input connections to TPU{3,3}, so we can use both connections simultaneously to double our ICI bandwidth.

ICI b/w = 9×10^{10} (bidirectional through 2 paths).

We can think of each path like a pipe.

We have $6 \cdot 1 \mu s$ startup time to fill the pipeline at which we reach steady state where data reaches by bytes/bw.

(i) Time for first byte to reach: $6 \cdot 1 \mu s = 6 \mu s$.

(ii) Total time:

$$6 \mu s + \frac{2 \cdot 8 \cdot 128 \cdot 8192}{9 \times 10^{10}} s = \underbrace{6 \mu s}_{\text{startup}} + \underbrace{186 \mu s}_{\text{steady state}} = 192 \mu s.$$

Q6

A : bf16[128 · 1024, 128 · 1024], split across TPU v5e 4 × 4 slice but offloaded to host DRAM.

Want:

1. Copy A to TPU{0,0}.
2. Multiply by vector bf16[8, 128 · 1024].

Time for step 1 — two options

Option 1. PCIe-copy into TPU HBM, then transfer across ICI. (Note: ICI transfers happen from HBM, no need to move into VMEM first.)

- PCIe load time:

$$T_{\text{PCIe}} = \frac{2 \cdot (128 \cdot 1024)^2}{16 \cdot 1.6 \times 10^{10}} = 0.07 \text{ s.}$$

- ICI transfer time. TPU{0,0} has 2 input sockets, so effective ICI bw = $2 \cdot 4.5 \times 10^{10}$ bytes/s. Neighbors are 1 hop away $\Rightarrow$ startup latency $1 \mu s$. Afterwards, $\frac{15}{16}(128 \cdot 1024)^2$ bytes come in:

$$T_{\text{comm}} = 1 \mu s + \frac{\frac{15}{16} (128 \cdot 1024)^2}{2 \cdot 4.5 \times 10^{10}} s \approx 0.17 \text{ s.}$$

- Total time for step 1 (Option 1): $T_{\text{comm}} + T_{\text{PCIe}} \approx 0.24 \text{ s.}$

Option 2. From host DRAM, transfer using DCN.

- DCN bandwidth = 6.25×10^9 bytes/s.
- Total time (also includes time to transfer from host DRAM to TPU HBM):

$$\frac{\frac{15}{16} \cdot (128 \cdot 1024)^2}{6.25 \times 10^9} s = 2.57 \text{ s.}$$

Option 1 is way better.

Time for step 2 — multiply by vector

$$T_{\text{mem}} = \frac{(128 \cdot 1024)^2 + 2 \cdot 8 \cdot 128 \cdot 1024 + 2 \cdot 8 \cdot 128 \cdot 1024}{8.2 \times 10^{11}} = 0.02 \text{ s}$$

$$T_{\text{comp}} = \frac{2 \cdot (128 \cdot 1024)^2 \cdot 8}{1.97 \times 10^{14}} = 0.001 \text{ s}$$

Bounds on total time

lower bound: $\max\{T_{\text{comm}}, T_{\text{mem}}, T_{\text{comp}}, T_{\text{PCIe}}\} \approx 0.17 \text{ s}$ (assuming perfect overlap)

upper bound: $T_{\text{comm}} + T_{\text{mem}} + T_{\text{comp}} + T_{\text{PCIe}} \approx 0.26 \text{ s}$

Appendix A — Pop Quiz

- TPU v5p has 2 cores.
- Each core's VPU has $4 \cdot 8 \cdot 128$ ALUs.

$$\text{TPU v5p vector FLOPs/s} = \underbrace{(2 \cdot 4 \cdot 128 \cdot 8)}_{\text{FLOPs}} \cdot \underbrace{1.75\text{e}9}_{1/\text{s}}$$

$$= (2 \cdot 4 \cdot 128 \cdot 8 \cdot 1.75\text{e}9) \text{ FLOPs/s} = 1.43 \times 10^{13} \text{ FLOPs/s.}$$